

# EE5203 L01 Practice Worksheet

*From C code to a blinking LED: what each step proves - Basri Erdogan*

**Time:** 30-45 minutes

**Rule:** Try every question before you open the solutions. Short answers are fine. One sentence is usually enough.

## Part A - Name the parts (5 minutes)

1. Write one word from this list next to each item: **processor, microcontroller, peripheral, board, embedded system.**

| Item                                                   | Your word |
|--------------------------------------------------------|-----------|
| Arm Cortex-M4                                          |           |
| STM32F401RE                                            |           |
| Nucleo-F401RE                                          |           |
| TIM2 (a timer inside the chip)                         |           |
| A thermostat: STM32 board + temperature sensor + relay |           |

2. An embedded system has limits. Give one example for each limit. Example for time: *“it must react within 10 ms”*.
- a. Memory: ...
  - b. Energy: ...
  - c. Safety: ...
3. The console says **“Build: 0 errors”**, but the LED does not blink. Is this possible? **Yes / No**. Explain in one sentence.

## Part B - The path from code to LED (10 minutes)

4. Put the six steps in order. Write 1 (first) to 6 (last).

| Step                                   | Order |
|----------------------------------------|-------|
| Flash the firmware into the chip       |       |
| Link the object files into one program |       |
| Edit <code>main.c</code>               |       |
| Compile the C files                    |       |
| Watch the LED                          |       |
| Look at a variable at a breakpoint     |       |

5. Each message comes from one step of the path. Match them. Steps: **compile, link, flash, run.**

| Message or symptom                                                        | Step |
|---------------------------------------------------------------------------|------|
| expected ';' before '}'                                                   |      |
| undefined reference to process_sample                                     |      |
| No ST-LINK detected                                                       |      |
| The program stops at the breakpoint, but the variable has the wrong value |      |

6. Each sentence below is a claim. Write **what you would look at** to prove it: the build console, the download message, the debugger, or the board itself.
- “My code compiled.”
  - “The program is inside the chip.”
  - “The loop runs again and again.”
  - “The LED blinks once per second.”

### Part C - Predict before you run (10 minutes)

7. Read the code. Do **not** run it.

```
1 int count = 0;
2
3 while (count < 3) {
4     count = count + 1;
5 }
```

- a. Fill the table.

| After loop pass | count is |
|-----------------|----------|
|-----------------|----------|

|   |  |
|---|--|
| 1 |  |
|---|--|

|   |  |
|---|--|
| 2 |  |
|---|--|

|   |  |
|---|--|
| 3 |  |
|---|--|

- b. What is the final value of `count`?
- c. On which line would you put a breakpoint to watch `count` change? Why there?
8. Two programs. Only one number is different.

|                                                                                           |                                                                                           |
|-------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| <pre>/* Program A */ while (1) {     HAL_GPIO_TogglePin(...);     HAL_Delay(500); }</pre> | <pre>/* Program B */ while (1) {     HAL_GPIO_TogglePin(...);     HAL_Delay(100); }</pre> |
|-------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|

- a. In 10 seconds, about how many times does the LED toggle in Program A? In Program B?

- b. What stays the same in both programs?
- c. How would you check your answer with a phone stopwatch?
9. You change 500 to 100, press **F5**, and the LED blinks exactly as before. Give **three** possible reasons. For each reason, write one quick test.

| Possible reason | Quick test |
|-----------------|------------|
| 1.              |            |
| 2.              |            |
| 3.              |            |

### Part D - Ask for help the right way (10 minutes)

10. A student says: “*My code does not work.*” Here are the facts: the build and the download succeeded; LD2 should blink every 0.5 s; LD2 stays off; a breakpoint inside `while (1)` is never hit. Write the four lines:
  - **Expected:** ...
  - **Observed:** ...
  - **Evidence:** ...
  - **Next test:** ...
11. A breakpoint on `HAL_GPIO_TogglePin(...)` is never reached. Number these checks from 1 (do first) to 5 (do last). Then explain your number 1 in one sentence.

| Check                                               | Order |
|-----------------------------------------------------|-------|
| Replace the USB cable                               |       |
| Check that the program reaches <code>main()</code>  |       |
| Check that the correct project was flashed          |       |
| Check whether the program is stopped somewhere else |       |
| Check the loop and <code>if</code> conditions       |       |

12. The LED blinks correctly. In the debugger, `toggle_count` still shows 0 after 10 seconds. Can the blinking LED prove that `toggle_count` is correct? **Yes / No**. Give two reasons.

### Part E - Your project idea (5 minutes)

13. Choose one device you know: an alarm clock, a kettle, a bike light, a parking sensor, anything. Fill the five boxes with one word or a short phrase each.

| Input | Acquisition | Processing | Output | Communication |
|-------|-------------|------------|--------|---------------|
|       |             |            |        |               |

14. Describe the **smallest version that works** in at most three sentences. Then name **one extra feature** you

would add only after the small version works.

## Self-check

Mark one:

- ☐ I am ready for Lab 1.
- ☐ I need to review the path from code to LED (Part B).
- ☐ I need help with C loops and variables (Part C).
- ☐ I need help with the VS Code / CubeMX setup.